



Enabling zero added-coalescent waterborne acrylic coatings with cellulose nanocrystals

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ABSTRACT

Waterborne acrylic coatings contain volatile and non-volatile coalescents. Volatile organic compound (VOC) coalescents enable film formation of hard acrylic polymers at ambient temperature, but their evaporation results in air pollution and health concerns. Non-volatile alternatives are available, but they remain in the coating long-term and as a result films are often tacky and lack adequate hardness. Reinforcing mineral fillers are available, but are associated with sustainability challenges inherent to mineral mining. An alternative approach demonstrated here utilizes a renewably-sourced, plant-based hardening filler dispersed in the water phase, with soft polymer latexes that don't require VOCs for film formation. We reinforced coalescent-free and ambient film-forming butyl acrylate/methyl methacrylate/methacrylic acid latexes with cellulose nanocrystals (CNCs) by a post-synthesis blending approach. CNCs possess high elastic modulus and tensile strength, making them promising candidates to improve the mechanical performance of soft ambient film-forming acrylic latexes. CNCs form aggregates within the interstices between latex particles as water dries, however film coalescence is not compromised. Films with 15 wt% CNC showed almost 230 % improvement in Koenig hardness and 10x improvement in the nanoindentation hardness compared to neat films, achieving properties similar to hard VOC-containing acrylic binders in a zero-added VOC formulation.

1. Introduction

Paints and coatings with a market of ~43 million tons in 2016 [1], have annual volatile organic compound (VOC) emissions of ~2 million tons [2]. Reducing VOC release has driven development of waterborne paints, but despite advances many interior waterborne products contain 250 g/L VOCs [3]. In addition to inherent health effects [4], VOCs react with atmospheric nitrogen oxides, forming ground-level ozone [5] that contributes to global warming and harms health [6]. Coating producers are under increasing pressure to decrease VOC content because of legislation and customer demand. However, transitioning to low VOC products brings a challenge known as the “film formation dilemma”, in which the VOC organic coalescents are needed to plasticize individual latex particles so that they form hard, durable coatings following VOC evaporation.

Acrylics are the most used binder resin in waterborne coatings.

Hardness of the acrylic is an important aspect of coating durability. Soft latex particles that easily coalesce to form a film lack adequate hardness. Latexes are often formulated with copolymers with high glass transition temperature (T_g) but VOC coalescents are added to temporarily plasticize the particles to enable film formation at ambient conditions [7]. Industry also employs non-evaporative plasticizers that provide coalescence. However, the film remains tacky after curing. Thus, alternative approaches are desired to achieve the performance of high T_g acrylic latex coatings in the lower VOC formulations of today.

One approach is to add a hardening filler to the polymer matrix. Mechanically robust nanoscale fillers have been shown to improve polymer mechanical performance [8], including nanoscale fillers such as clay [9], silica [10], and carbon nanotubes [11]. Cellulose nanocrystals (CNCs) are proposed as a renewable filler to enhance mechanical performance of soft latex films, due to their high crystallinity, strength, and modulus, low density (relative to mineral fillers), and optical transparency (when

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